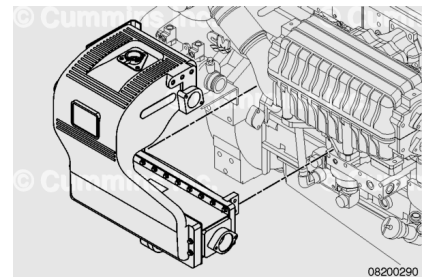


Trainings ▾

Install

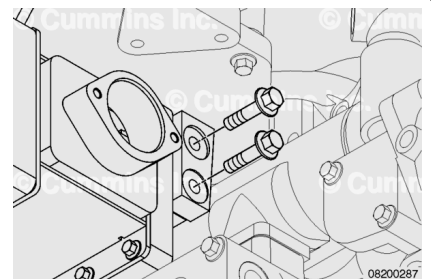
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

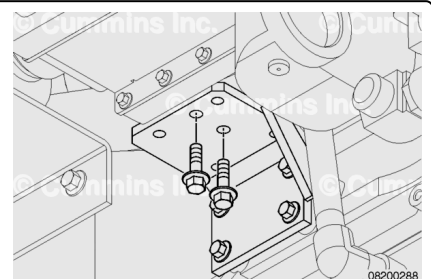


Install the heat exchanger and expansion tank assembly onto the engine.

Install but do **not** tighten the two mounting capscrews to the front of the heat exchanger.



Install the two mounting capscrews to the lower heat exchanger rear mounting bracket.

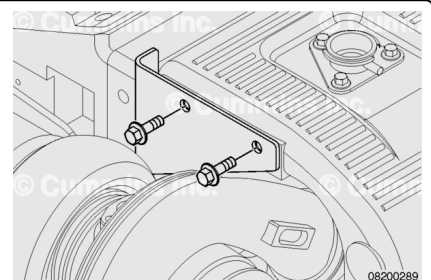


Install the two mounting capscrews to the upper heat exchanger mounting bracket.

Hand tighten all the capscrews once the heat exchanger assembly is in place.

Tighten the capscrews in the following sequence:

1. Front mounting capscrews
2. Lower mounting capscrews



3. Top mounting capscrews

Torque Value: 44 n•m [32 ft-lb]

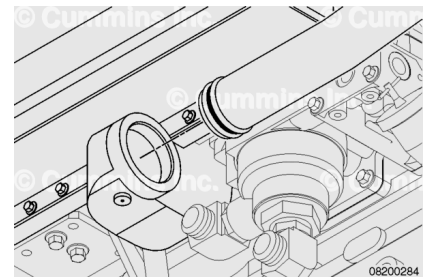
If the upper and lower mounting brackets were loosened to gain the heat exchanger alignment, tighten them to the following torque specifications.

Upper Bracket Capscrews	30 n.m	[22 ft-lb]
Lower Bracket Capscrews	44 n.m	[32 ft-lb]

Use Lubriplate™ paste on the o-rings and install them onto the end of the coolant tube and into the groove on the flange of the coolant suction tube.



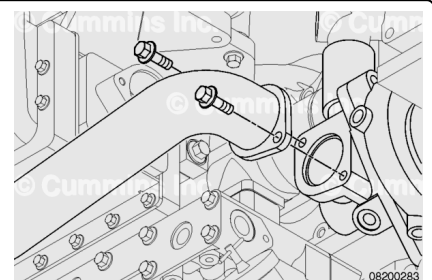
Insert the coolant tube with the o-ring into the heat exchanger outlet connection.



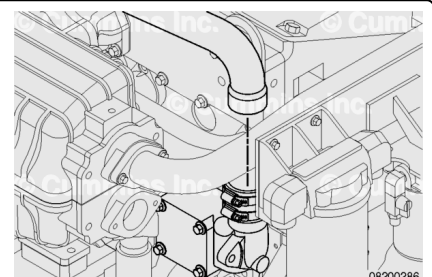
Verify the o-ring is still in place.

Install the flanged end of the tube and two capscrews to the coolant water pump inlet.

Torque Value: 44 n•m [32 ft-lb]



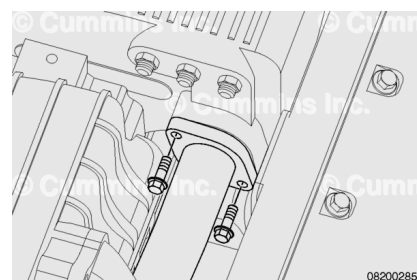
Insert the coolant return tube into the hose on the thermostat housing.



Use Lubriplate™ paste on the o-ring and install it in the groove on the coolant return tube flange.

Install the coolant return tube to the expansion tank.

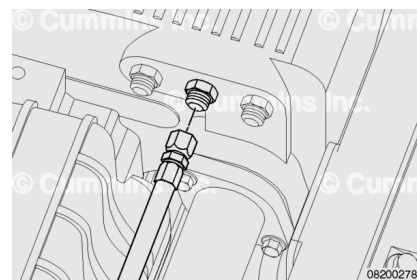
Tighten the hose clamps and capscrews.



Tube to Expansion Tank Capscrews	24 n.m	[212 in-lb]
Hose Clamps	8 n.m	[71 in-lb]

Install the coolant vent line.

Torque Value: 15 n•m [133 in-lb]

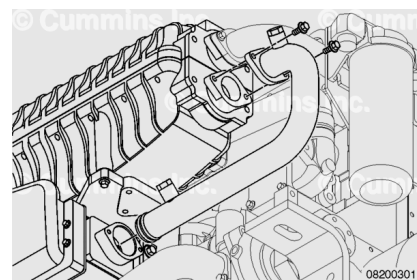


Use Lubriplate™ paste on the o-rings and install them on each end of the sea water tube.

Install the lower sea water tube.

Install the 2 capscrews to the flanged end on the sea water coolant tube to the aftercooler.

Tighten the capscrews evenly to be sure the flat face of the flange mates evenly to the aftercooler.

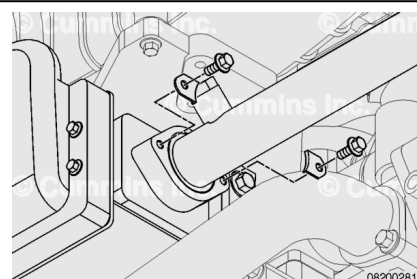


Sea Water Tube to Expansion tank Capscrews	24 n.m	[212 in-lb]
--	--------	-------------

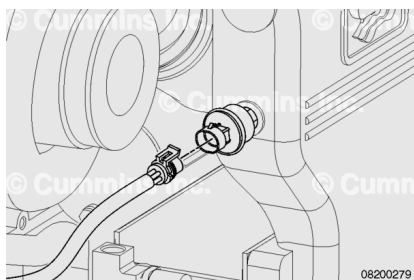
Install the two clamps and capscrews to both sides of the heat exchanger sea water inlet connection.

Tighten the capscrews.

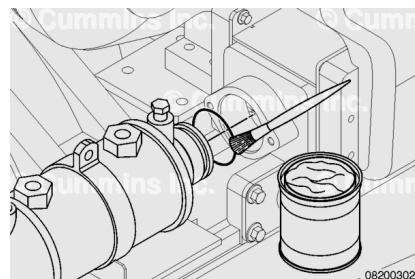
Sea Water Tube to Heat Exchanger Clamp Capscrews	24 n.m	[212 in-lb]
--	--------	-------------



Connect the engine wiring harness to the coolant level sensor.



Use Lubriplate™ paste on the o-ring and install it onto the marine gear oil cooler.



Insert the marine gear oil cooler (end with o-ring) into the heat exchanger core.

Insert the marine gear oil cooler in as far as possible, then back it out as needed to secure the bracket.

Install and tighten the 2 capscrews.

Torque Value: 44 n•m [32 ft-lb]

